

2025

COMPUTER SCIENCE — HONOURS

Paper : DSCC-6

(Microprocessor and its Applications)

Full Marks : 75

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer **any five** questions :

2×5

- (a) What is the purpose of READY signal with respect to microprocessor 8085?
- (b) Which operations are performed by the Arithmetic Logic Unit (ALU) of the 8085 Microprocessor?
- (c) How many hardware interrupts 8085 supports? Name the hardware interrupts.
- (d) What are opcode and operand?
- (e) Explain the software instructions EI and DI.
- (f) What is the function of ALE and how does it function?
- (g) Explain the function of the two DMA signals HOLD and HLDA.
- (h) What is the maximum addressable memory capacity of the 8085 microprocessor? Provide a justification based on its architectural constraints.

Answer **any three** questions.

- 2. Describe the address-data bus demultiplexing process in the 8085 Microprocessor with an appropriate diagram. 5
- 3. Describe how control signals are generated in the 8085 Microprocessor to enable separate access to memory and I/O devices. Draw suitable diagram. 5
- 4. What is the maximum number of input and output ports supported in the I/O mapped I/O scheme in 8085 Microprocessor? Describe the operations performed by the IN and OUT instructions using 8-bit addressing. 1+4
- 5. Compare I/O mapped I/O and memory mapped I/O in the context of the 8085 microprocessor, highlighting their key differences. 5
- 6. Differentiate between RAR and RRC instructions with examples. 5

Answer *any five* questions.

7. Draw the internal architecture of Microprocessor 8085 explaining each section in brief. 5+5
8. (a) What will be the content of Accumulator of microprocessor 8085 after the execution of XRA A? Justify your answer. 5+5
- (b) Are the JMP 16-bit address and PCHL instruction equivalent in functionality in the 8085 microprocessor? Justify your answer with an appropriate example. 2+3+5
- (c) What is stack? Explain the instruction related to stack operation. 5+5
9. What are the different general purpose and special purpose registers available in Microprocessor 8085? Explain each with suitable examples. 3+3+4
10. Define the term 'interrupt'. Distinguish between hardware and software interrupts in the context of the 8085 Microprocessor. Describe how the RST 5 instruction (Opcode – EFH) can be inserted onto the data bus using a suitable hardware implementation. 3+3+4
11. What are the different status flags in Microprocessor 8085? Explain each with suitable example. 3+7
12. Demonstrate the interfacing of the Programmable Peripheral Interface (PPI) 8255 with the 8085 Microprocessor using the I/O-mapped I/O method. Assign port addresses as specified : 6+4

Name of Port	Port Address	Remarks
Port-A	80H	Physical address of Port-A of 8255
Port-B	81H	Physical address of Port-B of 8255
Port-C	82H	Physical address of Port-C of 8255
CWR	83H	Physical address of CWR of 8255

13. Analyze the contents of registers after the execution of each instruction in the 8085 Assembly Language Program. *10

LXI H 3000H

MVI M, 32H

MOV B, M

XRA A

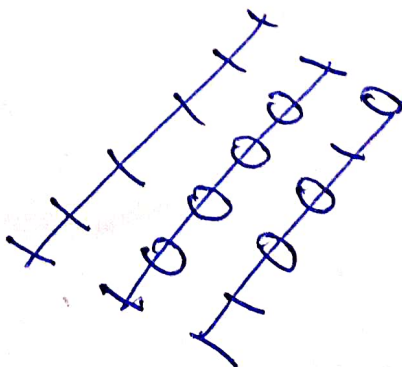
MOV A, B

ORA B

Back : DCR B

JNZ Back

HLT



(3)

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5+5

14. Write in short on the following (*any two*) :

- (a) BSR mode of PPI 8255
 - ~~(b)~~ Draw the timing diagram of Opcode fetch operation.
 - (c) Explain the instruction SHLD, 16-bit address and LDAX B with suitable example.
 - ~~(d)~~ Non-maskable Interrupt.
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